

ORACLE
AI World

Cut the cost, kill the complexity!

Agile Apps – Unified Data

Arman Ashouriha, Vodafone

Michael Doherty, Renesas

Jean-Pierre Dijcks, Oracle

Agenda

A conversation about data, architecture, capabilities, cost savings and delivering business value

- 1 Moving from Mongo to Oracle – The Vodafone Story
- 2 Building apps on Oracle JSON – The Renesas Story
- 3 Some Takeaways
- 4 Q&A

Simplify and Lower Cost

Inside Vodafone's switch from MongoDB to Oracle

Featured Speaker



Arman Ashouriha

Head of Oracle Cloud
Infrastructure Modernisation
Programme, Vodafone

With vast experience in managing international teams and programmes across Telecom and IT business sectors, Arman's professional experience includes technology & digital transformation, strategy consulting, service delivery across Consumer, Enterprise, Cloud and IoT services.



Vodafone

Leading European and African telecoms company providing mobile and fixed services to over 330 million customers in 15 countries

Vodafone's Oracle Cloud Modernization Programme



Architecture and Data Complexity



Overall cost and improved business benefits



Ability to do more with Oracle

Improving Data



Inside Renesas' quest to efficiently collect, process,
and store large amounts of data

Featured Speaker



Michael Doherty

Head of IT Infrastructure, Renesas

Mike is currently Head of Global IT Infrastructure at Renesas Electronics with strategic and operational responsibility for Data Centres, Cloud Infrastructure, Networks, Compute, Storage, Identity Management and Digital Workplace.

Mike is an experienced IT leader with a career spanning operations, delivery, consultancy and development. Mike started his career with roles in scientific research and has subsequently held positions across multiple other domains including finance, technology and telecoms, working for some of the largest organizations in these sectors.

Renesas Electronics Corporation is a global leader in the semiconductor industry, specializing in embedded semiconductor solutions

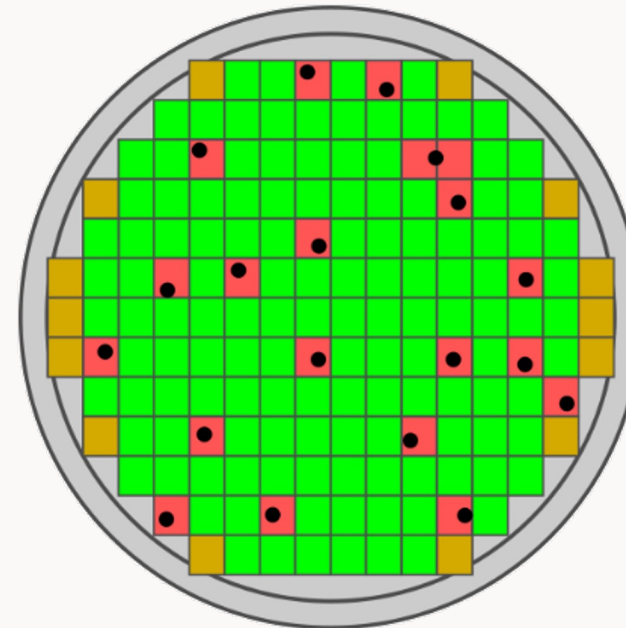


Renesas offers a broad product portfolio and comprehensive semiconductor solutions for four key segments:

- ✓ Automotive
- ✓ Industrial
- ✓ Infrastructure
- ✓ IoT (Internet of Things)

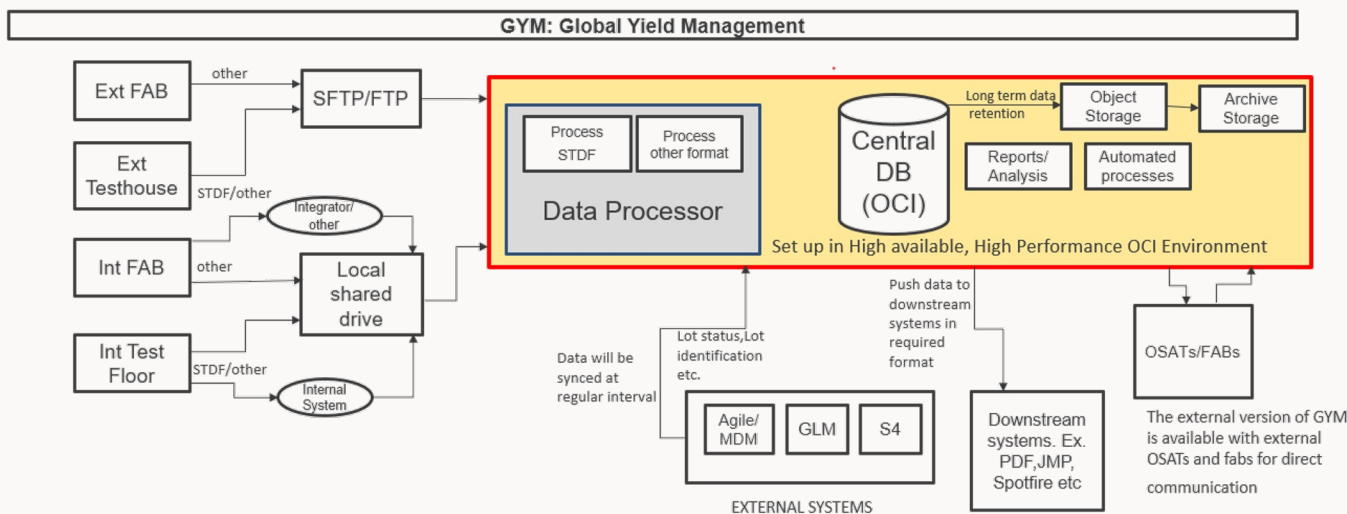
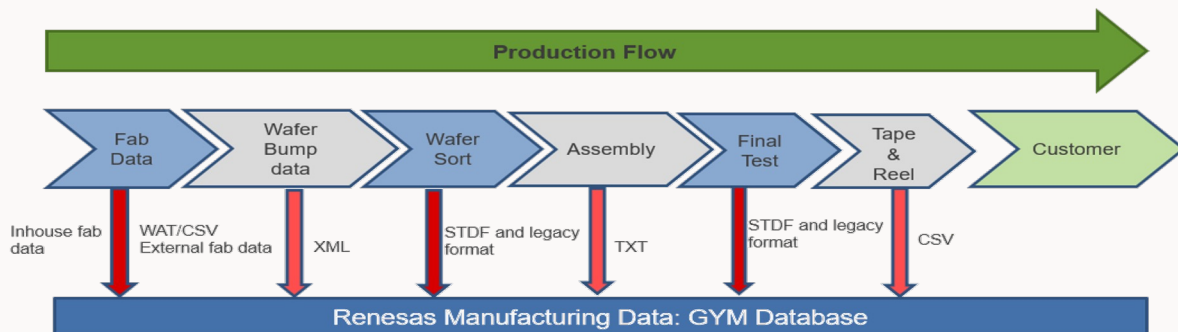
Product Category	Key Sub-Categories / Families
Microcontrollers (MCUs) & Microprocessors (MPUs)	RA Family, RX Family, RL78 Family, RH850 Family, R-Car Family
Analog Products	Amplifiers & Comparators, Data Converters, Interface ICs, Audio/Video Products
Power Management ICs (PMICs)	Voltage Regulators, Multi-Channel PMICs, Battery Management ICs, LED Driver ICs
ASIC & IP Products	ASIC Design, IP Cores & Licensing
Connectivity Solutions	Wireless, Timing & Synchronization, USB & Serial Connectivity
Memory and Logic ICs	SRAM, NAND & NOR Flash Memory, Logic Devices
Sensor Products	Optical Sensors, Temperature Sensors, Sensor Signal Condition

- Data collection from ATEs (Automated Test Equipment) for yield monitoring is a common practice in the semiconductor world
- So, we are constantly focusing on improving the process of data collection, data transfer and data storing
- Technology Roadmap:
 - Before end of this year: database will be upgraded to 23ai
 - We will convert Blob to native JSON early next year



- - defect
- - defective die
- - good die
- - partial edge die

DATA FLOW



- Global Yield Management (GYM) is the global system for yield related manufacturing data within Renesas for the entire production flow.
- It is set up in a **high available Oracle cloud environment** to provide service (24/7) to the users across functions and locations
- Previously, all perimetric data was stored in standard relational tables using rows and columns. Now, the test **data is stored in JSON format**, where the individual test results are organized as array elements.

Renesas' Business Benefits



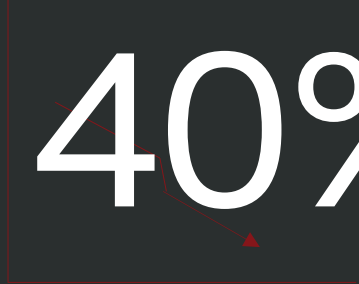
A line graph showing a 30% reduction in storage. The y-axis is labeled '30%' and the x-axis is labeled 'consumed before compression'. A blue line starts at a high point and drops to a lower point, with a green arrow pointing down to the lower point.

30% less storage
consumed before
compression



A line graph showing a 3x speedup in queries. The y-axis is labeled '3x' and the x-axis is labeled 'of queries at minimum'. A blue line starts at a high point and drops to a lower point, with a blue arrow pointing down to the lower point.

3x speedup
of queries at
minimum



A line graph showing a 40% reduction in operational cost. The y-axis is labeled '40%' and the x-axis is labeled 'operational cost'. A red line starts at a high point and drops to a lower point, with a red arrow pointing down to the lower point.

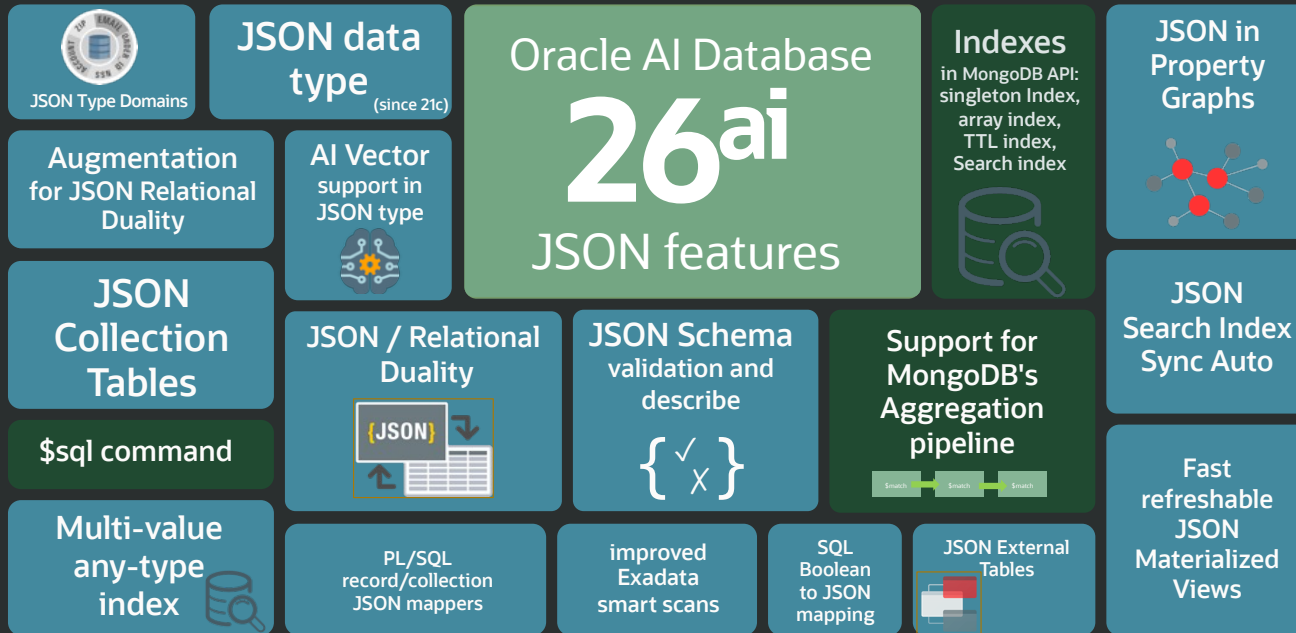
40% lower
operational cost

Recap

An abstract graphic on the right side of the slide. It features several overlapping organic shapes. At the top right is a purple shape with a white concentric line pattern. Below it is a large blue shape. To the right of the blue shape is a green shape with a white pattern of overlapping circles. At the bottom right is a red shape. A small white circle is located in the bottom right corner of the red shape.

The Oracle Products supporting successful customers like Vodafone and Renesas

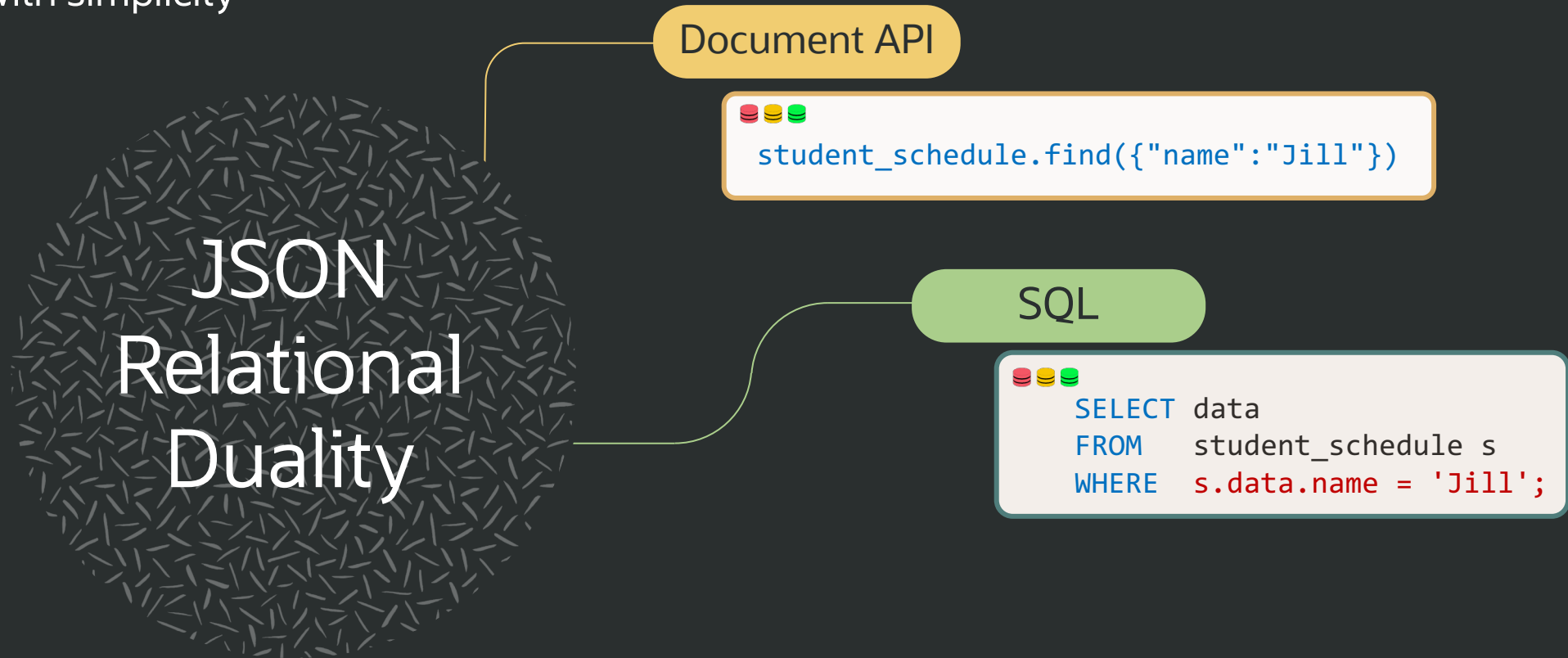
Oracle JSON Database – Powering Worldwide Business Success



Oracle JSON Database delivers key document store capabilities:

- A full range of native structures and support for JSON documents
- The ability to run multiple workloads (SQL Analytics, Graph, Spatial, AI, and more) on the same data. No data movement required!
- Instant integration with the full gambit of Oracle, including security, governance, scalability, and performance features
- An API to switch from various document databases without risk

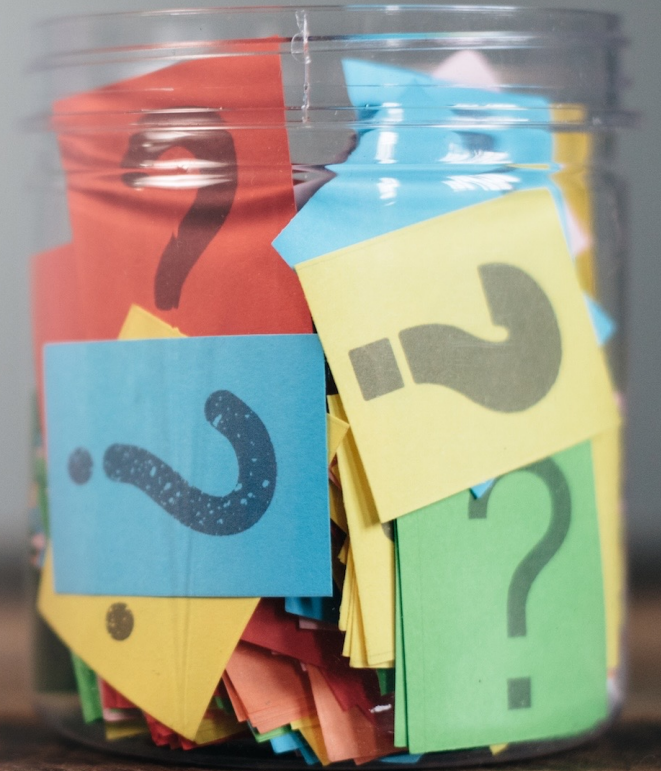
A **breakthrough technology** to help developers integrate the same data across multiple interfaces serving many use cases with simplicity



Create JSON interfaces on existing relational data and enable new workloads and integrations without duplicating data

Develop new applications on a single optimized data tier, enable Java Script and SQL (+ more) for reads and writes through both paths

Q&A



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